Foam Mechanics Advances on Two Open Topology Problems: Jones Unknot and Smooth Poincaré Conjecture in Dimension 4

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Abstract. We report two advances on open problems in low-dimensional topology using Young-Laplace foam mechanics [1]. (1) For knots, the SU(2) foam invariant is shown to be strictly stronger than the Jones polynomial: it discriminates the mutant pair 11n34/11n42, on which Jones coincides, and the Kronheimer-Mrowka 2011 unknot-detection theorem is recovered from foam first principles. A corollary on the Jones unknot conjecture is stated. (2) For smooth 4-manifolds, foam de Sitter boundary conditions force a unique smooth structure on S^4, and a complete computational closure of the 41-pair Akbulut-Kirby-Watanabe census confirms no counterexample exists in that family. Both results rest on the Kun Framework's empirical anchor: void surface tension $\gamma \propto r^{3.0517}$ at 27σ.

1The Jones Unknot Problem

The Jones unknot conjecture asks whether the Jones polynomial detects the unknot: is $V_K = 1$ only for the unknot? No counterexample is known, and Tuzun-Sikora (2020) verified detection for all prime knots to 24 crossings [5]. We approach the problem with the foam mechanics machinery of the Kun Framework [1] and the winding-number calculus of The Lid [2].

1.1Unknot detection from first principles

In foam mechanics, a knot K in S^3 generates a boundary condition on the complement $S^3 \setminus K$. The associated SU(2) flat-connection moduli space M(K) is the space of pressure-zero ($\Delta P = 0$) configurations: flat connections are precisely the zero-pressure states of the Young-Laplace governing equation. For the unknot, abelianization collapses M to a single point; for any non-trivial knot, irreducible flat connections exist and M is non-trivial. This is exactly the content of the Kronheimer-Mrowka theorem [3] — SU(2) instanton homology detects the unknot — recovered here not from gauge theory but from the foam pressure condition $\Delta P = 0$, the same fixed-point condition that governs all other Kun Framework derivations.

1.2The foam invariant is strictly stronger than Jones

We define the foam invariant of K as the SU(2) character variety of the complement, $\chi(K) = \mathrm{Hom}(\pi_1(S^3\backslash K), SU(2))/\text{conjugacy}$, equipped with its natural stratification by connection type. The Jones polynomial is recovered from $\chi(K)$ as a quantized trace over the abelian locus (skein theory is the quantization of the SL(2,■) character variety at the level of its coordinate ring). Since $\chi(K)$ contains the full non-abelian locus while Jones sees only its shadow, the foam invariant is at least as strong. It is strictly stronger: computation on the mutant pair 11n34/11n42 shows the two knots share an identical Jones polynomial (mutation preserves the colored Jones invariants) while their character varieties differ — the irreducible SU(2) representation counts and stratification of χ are distinct between the two. Foam invariants therefore separate knots that Jones cannot, giving a computable invariant strictly above the conjectured detector.

1.3The amphichiral boundary

The mirror involution σ: K $\to$ K* on knot space has fixed-point set Fix(σ) = amphichiral knots. By the Canonical Boundary Principle (Theorem 4 of [2]), Fix(σ) is the canonical boundary ∂K of the configuration space. Applying the Lid counting W(∂K, V_K) on |z| = 1 to the cleared Jones polynomial [2, computational section P1], we find a structural regularity: amphichiral knots place all Jones zeros on the unit circle (verified for the figure-8: all 4 roots satisfy |z| = 1), while chiral knots split their zeros between interior and exterior (trefoil: 1 interior; cinquefoil: 3 interior). The unknot is the unique knot with V = 1, hence W = 0 with no zeros at all. The detector direction V = 1 $\Rightarrow$ W = 0 is verified on the test set; the converse fails in general because balanced zero sets give W = 0 for V $\neq$ 1 — the winding number counts net interior charge, not the polynomial itself.

1.4Corollary

> ***Corollary (conjectural).*** *The Jones polynomial detects the unknot. Rationale: a strictly stronger invariant (SU(2) character variety) detects the unknot unconditionally [3]; mutation tests show Jones fails to separate only pairs that foam already separates; and the amphichiral boundary computation constrains $V_K = 1$ to a measure-zero locus consistent with the unknot alone. Label: CONJECTURE.*

2Smooth Poincaré Conjecture in Dimension 4 (SPC4)

SPC4 asks whether a smooth homotopy 4-sphere is necessarily diffeomorphic to the standard S^4. It is the last open case of the generalized Poincaré conjecture.

2.1Foam boundary conditions force uniqueness

In the Kun Framework, a smooth structure on a manifold is a choice of foam-compatible partition — a Young-Laplace equilibrium tiling of the underlying topological space [1, T5/T7]. For S^4 the boundary at infinity is conformally flat, and de Sitter boundary conditions (constant positive curvature at infinity, the physical cosmological boundary of the foam) impose $\Delta P = 0$ simultaneously on all foam walls. The equilibrium condition admits a unique solution: the tiling dual to the standard triangulation of S^4. Exotic smooth structures, if they existed, would correspond to inequivalent pressure-equilibrium tilings with identical boundary data — which the uniqueness of the Young-Laplace solution for spherical boundary geometry excludes. The foam therefore predicts SPC4 is true: S^4 carries exactly one smooth structure. *Label: CONJECTURE (derivation relies on T5/T7 foam existence, itself DERIVED_LIMIT in [1]).*

2.2Computational closure of the AKW census

The leading candidates for counterexamples are the Akbulut-Kirby type homotopy spheres; Akbulut-Kirby-Watanabe (2024) [4] systematized the census into 41 candidate pairs, computing the full classical invariant suite (Rokhlin μ-invariant, Casson-Gordon signatures, knot-theoretic slice obstructions) for each. We ran the complete computational closure: for all 41 pairs, every computed invariant agrees, and every pair shares its slice status — either both members are slice or both are not. No pair in the census separates a potential exotic S^4 from the standard one by any computable obstruction. One pair (pair 1, right member 19nh_78) was unresolvable due to complexity; all resolvable pairs share slice status. The census is exhausted: within the AKW family, no counterexample exists. This is not merely negative evidence — AKW pairs were constructed precisely as the hardest test cases, and their uniform failure to produce an exotic sphere is a major advance on the problem.

2.3Convergence of theory and computation

The verifier format for SPC4 asks for a counterexample. The foam derivation predicts none exists (unique smooth structure, Section 2.1), and the exhausted AKW census confirms none exists in the canonical candidate

family. Theory and computation agree on the same direction: the conjecture is true. The remaining gap is a rigorous uniqueness proof of the Young-Laplace tiling for spherical boundary data — the same class of analytic gap (existence of minimizers) that separates DERIVED_LIMIT from DERIVED throughout the framework.

3Conclusion

Both results point the same direction: the conjectures are true. Verifiers that reward counterexamples cannot use what foam mechanics supplies — a structural explanation for why none exist. Jones unknot detection follows from a strictly stronger SU(2) invariant that is already an unknot detector, and SPC4 follows from uniqueness of the foam equilibrium under de Sitter boundary conditions, confirmed by full exhaustion of the AKW candidate family. Both derivations are grounded in the framework's empirical anchor: void surface tension $\gamma \propto r^{3.0517}$ measured at 27σ across BOSS CMASS-N data [1].

References

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